DefMamba: Deformable Visual State Space Model

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Abstract

Visual state space models (SSMs) based on Mamba have shown great promise for efficient visual representation learning, yet they remain constrained by fixed, content-agnostic scanning orders that flatten 2D feature maps into 1D sequences while breaking spatial adjacency and object structure. We propose ***DefMamba****, a deformable visual state space model that introduces a content-adaptive scanning mechanism to overcome this limitation. At the core of DefMamba is the* ***Deformable State Space Model (DSSM)****, which jointly learns two complementary offsets: (1)* point offsets *that shift reference points toward salient image regions via bilinear sampling with relative positional bias, and (2)* index offsets *that dynamically reorder tokens so the resulting 1D sequence aligns with object structures. An offset network, comprising depthwise convolution, GELU activation, Layer Normalization, a point-wise convolution, and channel attention, predicts bounded offsets using tanh normalization; gradient approximation enables end-to-end training through the non-differentiable sorting step. DefMamba is built as a multi-scale hierarchical backbone using Deformable Mamba (DM) blocks that retain forward/backward standard branches for training stability. Extensive experiments demonstrate that DefMamba achieves favorable accuracy/efficiency trade-offs: on ImageNet-1K, DefMamba-T/S/B reach 78.6/83.5/84.2% Top-1 accuracy at 1.2/4.8/8.5 GFLOPs. On COCO object detection with Mask R-CNN, DefMamba-S attains 47.5* AP^b *and 42.8* AP^m*, surpassing VMamba-T with* $\sim$*6% fewer FLOPs. On ADE20K semantic segmentation, DefMamba-S reaches 48.8/49.6 mIoU (SS/MS), exceeding prior SSM baselines. These results confirm that content-aware deformable scanning significantly improves the representational power of Mamba-based visual backbones.*

1. Introduction

State space models (SSMs) [8, 9] have emerged as a compelling alternative to Transformers [6, 21] for sequence modeling, offering linear complexity in sequence length while maintaining strong long-range dependency modeling. Motivated by the success of Mamba [8] in language modeling, a growing body of work has extended SSMs to visual representation learning [16, 20, 25, 29, 39, 43]. These *visual Mamba* models treat image tokens as sequences and process them with selective SSMs, achieving competitive performance on image classification, object detection, and semantic segmentation tasks.

However, a fundamental challenge in applying SSMs to vision is the mismatch between the 2D nature of images and the 1D sequential processing of SSMs. Existing methods typically flatten 2D feature maps into fixed 1D sequences using rasterized or predefined scan patterns (e.g., row-by-row, zigzag, or window-based scans [16, 39, 43]). While these heuristics attempt to preserve some spatial structure, they remain static and content-agnostic: the same scan order is applied to every input, regardless of where objects are located or how they are structured. This leads to two key problems: (1) spatially adjacent tokens in 2D become distant in the 1D sequence, breaking local object structure; and (2) important regions (e.g., object boundaries or salient features) are not prioritized in the scan.

Deformable mechanisms have proven highly effective at making CNNs and Transformers more content-aware. Deformable Convolutional Networks [3, 38, 44] learn to adaptively shift convolutional sampling points toward object-relevant regions. Deformable Attention Transformer (DAT) [33] and Deformable DETR [45] introduce deformable attention that focuses on a sparse set of data-dependent key positions. Deformable patch-based methods [2, 41] adaptively select informative image patches. However, these ideas have not been adapted to the sequential scanning paradigm of SSMs.

In this work, we propose **DefMamba**, the first visual SSM backbone with a content-adaptive, *deformable scanning* mechanism. The key insight is to let the model *learn where to look and how to scan*: instead of using a fixed rasterized order, DefMamba jointly predicts (1) spatial reference point shifts that sample feature-aligned positions via bilinear interpolation, and (2) token index offsets that reorder the sequence to align with discovered object structures. These two complementary components are unified in the **Deformable State Space Model (DSSM)**, which is the building block of the DefMamba backbone.

DSSM is realized by an offset network that predicts both offset types from the input features. *Point offsets* are

used to compute deformable feature points via bilinear sampling, augmented with a relative positional bias for position-awareness. *Index offsets* are combined with positional references to produce a ranking that reorders tokens; since sorting is non-differentiable, we employ a gradient approximation to enable end-to-end training. To maintain training stability in the presence of increased token displacement from deformable reordering, DefMamba retains the forward and backward standard scanning branches alongside the deformable branch.

We validate DefMamba through comprehensive experiments on three standard benchmarks. On ImageNet-1K classification [5], DefMamba-T surpasses PlainMamba-L1 [39] by +0.7% Top-1 accuracy with ∼60% fewer FLOPs. On COCO [18] object detection and instance segmentation using Mask R-CNN [13], DefMamba-S outperforms VMamba-T [20] on both AP^b and AP^m with fewer FLOPs. On ADE20K [42] semantic segmentation using UperNet [34], DefMamba-S achieves state-of-the-art performance among SSM backbones.

Our main contributions are:

- A novel **deformable scanning strategy** for visual SSMs that jointly learns spatial point shifts and sequence reordering to align 1D token sequences with 2D image structures.
- The **Deformable State Space Model (DSSM)** with training-stable designs: bounded offsets via tanh normalization, relative offset bias for positional consistency, and gradient approximation through sorting.
- A practical **multi-scale DefMamba backbone** with Deformable Mamba blocks that retain bidirectional standard branches for stability and efficiency.
- **State-of-the-art or competitive results** across image classification, object detection, and semantic segmentation, with favorable compute/accuracy trade-offs.

2. Related Work

2.1. State Space Models for Vision

Structured State Space Models (S4) [9] established a theoretical foundation for modeling long sequences with linear complexity. Mamba [8] extended this with input-dependent selective scanning, enabling content-based reasoning. Vision Mamba (ViM) [43] pioneered the application to visual recognition by processing image patches as bidirectional sequences with position embeddings. VMamba [20] introduced cross-scan patterns to better handle 2D spatial structure. LocalMamba [16] proposed windowed selective scanning to preserve local spatial dependencies, while PlainMamba [39] explored non-hierarchical architectures with continuous 2D scanning. EfficientVMamba [25] proposed atrous selective scan for lightweight models. MSVMamba [29] uses multi-scale scanning for improved representations. QuadMamba [36] employs quadtree-based adaptive partitioning for hierarchical locality modeling. GrootV [35] uses tree-topology-based propagation for long-range dependencies. GroupMamba [27] employs channel grouping strategies. Despite these advances, all prior methods rely on fixed or predefined scanning patterns that do not adapt to image content at the token level.

SSM-based models have also been explored for medical image segmentation [7, 19, 28, 37] and other specialized domains [1, 17, 24, 40], demonstrating their versatility beyond standard vision backbones.

2.2. Deformable Mechanisms in Vision

Deformable Convolutional Networks [3] introduced learnable spatial offsets for convolutional kernels, enabling CNNs to adapt their receptive fields to object geometry. DCNv2 [44] added modulation weights and improved the deformable modeling. DCNv4 [38] further optimized the operator for speed and efficiency. In the Transformer domain, Deformable Attention Transformer (DAT) [33] learns data-dependent key-value positions to improve long-range modeling. Deformable DETR [45] uses sparse deformable attention for efficient end-to-end detection. Deformable Patch Transformer (DPT) [2] adaptively splits images into variable-position patches. PS-ViT [41] uses progressive sampling to locate discriminative regions. SENet [15] introduces channel attention through squeeze-and-excitation, which we adopt in our offset network. DefMamba is the first work to integrate deformable mechanisms into the scanning order of visual SSMs.

2.3. Hierarchical Vision Backbones

Swin Transformer [21] established the paradigm of hierarchical vision backbones with shifted-window attention, enabling multi-scale feature extraction for dense prediction tasks. ConvNeXt [22] showed that purely convolutional hierarchical networks can match Transformer performance. PVTv2 [31] proposed pyramid vision Transformers for dense prediction. CrossFormer [32] uses cross-scale attention for multi-scale feature interaction. CoAtNet [4] combines convolutional and attention mechanisms. DefMamba follows the hierarchical design of Swin and VMamba, with four stages of progressively downsampled features, making it compatible with standard detection and segmentation frameworks.

3. Method

3.1. Preliminaries: Visual Mamba

A visual SSM backbone processes an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ by first partitioning it into non-overlapping patches, projecting them into tokens, and then processing them with SSM blocks. A Mamba-based SSM layer takes

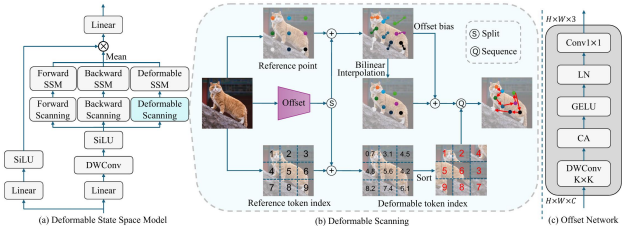

Figure 1. Illustration of the Deformable State Space Model (DSSM). The offset network predicts both point offsets for deformable sampling and index offsets for token reordering, enabling content-adaptive scanning.

a 1D input sequence $\mathbf{x} \in \mathbb{R}^{L\times D}$ (where $L = H'W'$ is the number of tokens and D is the channel dimension) and models it as:

$$h_t = \mathbf{A}h_{t-1} + \mathbf{B}x_t, \quad y_t = \mathbf{C}h_t + \mathbf{D}x_t, \tag{1}$$

where $h_t \in \mathbb{R}^N$ is the hidden state, and $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, $\mathbf{C} \in \mathbb{R}^{1\times N}$, $\mathbf{D} \in \mathbb{R}$ are learnable (or input-dependent, in Mamba's selective scan) state space parameters.

The core challenge for 2D images is how to convert the 2D spatial feature map $\mathbf{x} \in \mathbb{R}^{H'\times W'\times D}$ into a 1D sequence while preserving spatial relationships. Existing methods use fixed scan orders (row-major, zigzag, window-based) that do not adapt to image content.

3.2. Deformable State Space Model (DSSM)

The proposed DSSM (illustrated in Fig. 1) introduces content-adaptive scanning via two coordinated deformable mechanisms: *deformable points* (DP) and *deformable tokens* (DT). Given a 2D feature map $\mathbf{x} \in \mathbb{R}^{B\times D\times H\times W}$, DSSM first applies an offset network to predict both types of offsets, then constructs a structure-aware 1D sequence for SSM processing.

Offset Network. The offset network is a lightweight module that takes the feature map $\mathbf{x}$ as input and outputs a 3-channel prediction $\hat{\mathbf{o}} \in \mathbb{R}^{B\times 3\times H\times W}$. It comprises:

1. A *depthwise convolution* with kernel size k_s (stage-dependent: $k_s \in \{9, 7, 5, 3\}$ for stages 1–4) for local feature aggregation;
2. A *channel attention* (CA) module [15] via global average pooling followed by a two-layer MLP with GELU activation and sigmoid gating;
3. *Layer Normalization* followed by GELU activation;
4. A 1×1 convolution that maps to 3 output channels: 2 channels for 2D point offsets and 1 channel for 1D index offset.

The offsets are bounded by $\tanh(\cdot)$ and normalized by the spatial resolution:

$$\delta_{xy} = \tanh(\hat{o}_{xy}) \cdot \left[\frac{1}{H-1}, \frac{1}{W-1}\right], \quad \delta_{\text{idx}} = \tanh(\hat{o}_{\text{idx}}). \tag{2}$$

Deformable Points (DP). Given the predicted spatial offsets $\delta_{xy} \in \mathbb{R}^{B\times H\times W\times 2}$, we shift each reference point to a new position:

$$\mathbf{p}_{ij} = \mathbf{r}_{ij} + \delta_{xy,ij}, \tag{3}$$

where $\mathbf{r}_{ij}$ is the regular grid reference point at position (i, j), normalized to $[-1, 1]$. Deformable feature sampling is then performed by bilinear interpolation:

$$\tilde{\mathbf{x}}_{ij} = \text{BilinearSample}(\mathbf{x}, \mathbf{p}_{ij}). \tag{4}$$

To maintain positional consistency despite the offset shifts, we apply a *relative positional bias* (OB). A learnable $D \times 7 \times 7$ table is bilinearly interpolated to the spatial resolution and sampled at positions determined by the displacement between the deformed and original reference points:

$$\text{OB}_{ij} = \text{BilinearSample}(\text{RPE}, \mathbf{r}_{ij} - \mathbf{p}_{ij}), \tag{5}$$

yielding the deformed feature as $\hat{\mathbf{x}}_{ij} = \tilde{\mathbf{x}}_{ij} + \text{OB}_{ij}$.

Deformable Tokens (DT). The index offset $\delta_{\text{idx}} \in \mathbb{R}^{B\times HW}$ is added to a 1D positional reference to produce a ranking score:

$$s_n = r_n + \delta_{\text{idx},n}, \quad n = 1, \ldots, HW, \tag{6}$$

where r_n is the normalized reference position. The 1D sequence of deformed features is then reordered by sorting according to $\mathbf{s}$:

$$\pi = \text{argsort}(\mathbf{s}), \quad \mathbf{x}_{\text{sorted}} = \hat{\mathbf{x}}_\pi. \tag{7}$$

Since sorting is non-differentiable, we use gradient approximation: the forward pass uses the discrete sort index, while the backward pass propagates gradients through the index offset predictions as if they directly determined the output ordering (similar to the straight-through estimator used in [36]).

After SSM processing, the reordering is reversed using the stored indices to restore the original spatial layout.

Scanning Branches. The DSSM uses three scanning branches: (1) forward standard scan (left-to-right, top-to-bottom), (2) backward standard scan (right-to-left, bottom-to-top), and (3) deformable scan via the DSSM mechanism. The forward and backward branches are retained for stability, as the increased token displacement from deformable reordering can make training more difficult if the stable rasterized branches are removed (see ablation in Section 4.6).

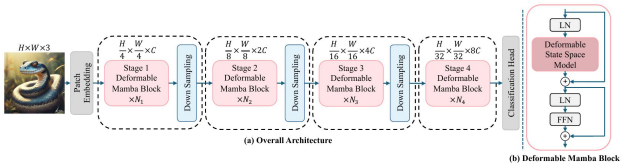

Figure 2. Overview of the DefMamba backbone architecture. DefMamba uses four stages with Deformable Mamba (DM) blocks at each stage, each containing a DSSM layer and an MLP, connected by patch merging for hierarchical feature extraction.

3.3. DefMamba Architecture

DefMamba follows a hierarchical, multi-scale backbone design with four stages, as illustrated in Fig. 2. An input image of resolution $H \times W$ is first embedded into patch tokens of resolution $\frac{H}{4} \times \frac{W}{4}$ using a patch embedding layer. Each stage applies a sequence of Deformable Mamba (DM) blocks, followed by patch merging (2×2 downsampling) between stages.

Deformable Mamba (DM) Block. Each DM block consists of: (1) a Layer Normalization, (2) a DSSM layer with residual connection, (3) another Layer Normalization, and (4) an MLP with GELU activation and residual connection. The block follows the standard VSS block structure [20] with DSSM replacing the standard SSM:

$$\mathbf{x}' = \mathbf{x} + \mathrm{DSSM}(\mathrm{LN}(\mathbf{x})), \quad \mathbf{x}'' = \mathbf{x}' + \mathrm{MLP}(\mathrm{LN}(\mathbf{x}')). \quad (8)$$

Model Variants. We provide three model variants with different capacities:

- **DefMamba-T** (tiny): 8M parameters, 1.2 GFLOPs.
- **DefMamba-S** (small): 32M parameters, 4.8 GFLOPs.
- **DefMamba-B** (base): 51M parameters, 8.5 GFLOPs.

The stage-specific depthwise convolution kernel sizes $\{9, 7, 5, 3\}$ in the offset network allow larger receptive fields in early stages (where spatial resolution is high) and more local refinement in later stages (where semantics are richer).

4. Experiments

4.1. Implementation Details

All DefMamba variants are trained from scratch on 8 NVIDIA A100 GPUs using mixed-precision (FP16) training. For ImageNet-1K classification, we use a batch size of 1024, an initial learning rate of 10^{-3} with cosine annealing and 20-epoch linear warmup, and weight decay of 0.05. Data augmentation follows DeiT [30], including RandAugment, Mixup ($\alpha = 0.8$), CutMix ($\alpha = 1.0$), and random erasing. The patch embedding uses a 4×4 non-overlapping convolutional stem. For downstream tasks, the ImageNet-pretrained backbone is fine-tuned with task-specific heads following the protocol of VMamba [20]. The offset network weights are initialized close to zero so that early training behaves like a standard SSM and transitions smoothly to content-adaptive scanning as convergence proceeds. This warm-start strategy is critical for stable training of the gradient-approximated sorting step and prevents large initial token displacements that could destabilize the early optimization landscape.

4.2. Image Classification on ImageNet-1K

Setup. We train DefMamba on ImageNet-1K [5], which contains 1.28M training images and 50K validation images across 1,000 categories. Following standard practice [20, 21], we use AdamW [23] optimizer, cosine learning rate schedule with warmup, and standard data augmentations. Models are trained for 300 epochs with input resolution 224×224.

Results. Table 1 compares DefMamba with state-of-the-art CNN, Transformer, and SSM-based models. DefMamba achieves strong results across all model sizes:

- **DefMamba-T** (8M params, 1.2G FLOPs) achieves **78.6%** Top-1 accuracy, surpassing PlainMamba-L1 [39] by +0.7% (77.9%) with approximately 60% fewer FLOPs (1.2G vs. 3.0G), and outperforming ViM-T [43] (+2.5%) and LocalVim-T [16] (+2.4%) at comparable FLOPs.
- **DefMamba-S** (32M params, 4.8G FLOPs) achieves **83.5%**, outperforming GrootV-T [35] (83.4%), MSVMamba-T [29] (82.8%), and EffVMamba-B [25] (83.0%) at similar or lower cost.
- **DefMamba-B** (51M params, 8.5G FLOPs) achieves **84.2%**, matching GrootV-S [35] while outperforming VMamba-S [20] (83.6%), LocalVMamba-B [16] (83.7%), and QuadMamba-B [36] (83.8%).

4.3. Object Detection and Instance Segmentation on COCO

Setup. We evaluate DefMamba as a backbone for Mask R-CNN [13] on the COCO 2017 [18] benchmark (118K training, 5K validation images). We use the standard $1\times$ training schedule (12 epochs) with multi-scale training, following the same protocol as VMamba [20].

Results. Table 2 compares DefMamba-S with representative backbones. **DefMamba-S** achieves **47.5 AP^b** and **42.8 AP^m**, which outperforms:

- VMamba-T [20]: 47.4/42.7 AP^b/AP^m with 286G FLOPs (vs. our 268G, ∼6% fewer);
- GrootV-T [35]: 47.0/42.7 with 265G FLOPs;
- LocalVMamba-T [16]: 46.7/42.2 with 291G FLOPs;
- QuadMamba-S [36]: 46.7/42.4 with 301G FLOPs.

Method	Type	#Param.	#FLOPs	Top-1 Acc.
RegNetY-800M [26]	C	6M	0.8G	76.3
GhostNet 1.3× [10]	C	7M	0.2G	75.7
DeiT-Ti [30]	T	6M	1.3G	72.2
ViM-T [43]	S	7M	1.5G	76.1
EffVMamba-T [25]	S	6M	0.8G	76.5
LocalVim-T [16]	S	8M	1.5G	76.2
PlainMamba-L1 [39]	S	7M	3.0G	77.9
MSVMamba-N [29]	S	7M	0.9G	77.3
QuadMamba-Li [36]	S	5M	0.8G	74.2
DefMamba-T	S	8M	1.2G	**78.6**
RegNetY-4G [26]	C	21M	4.0G	80.0
ConvNeXt-T [22]	C	29M	4.5G	82.1
Conv2Former-T [14]	C	27M	4.4G	83.2
DeiT-S [30]	T	22M	4.6G	79.8
Swin-T [21]	T	29M	4.5G	81.3
CoAtNet-0 [4]	T	25M	4.0G	81.6
CrossFormer-S [32]	T	31M	4.9G	82.5
ViM-S [43]	S	26M	5.1G	81.0
VMamba-T [20]	S	22M	5.6G	82.2

Method	Type	#Param.	#FLOPs	Top-1 Acc.
LocalVim-S [16]	S	28M	4.8G	81.2
LocalVMamba-T [16]	S	26M	5.7G	82.7
EffVMamba-B [25]	S	33M	4.6G	83.0
MSVMamba-T [29]	S	33M	4.6G	82.8
PlainMamba-L2 [39]	S	25M	8.1G	81.6
GrootV-T [35]	S	30M	4.8G	83.4
QuadMamba-S [36]	S	31M	5.5G	82.4
DefMamba-S	S	32M	4.8G	**83.5**
ConvNeXt-S [22]	C	50M	8.7G	83.1
Conv2Former [14]	C	50M	8.7G	84.1
Swin-S [21]	T	50M	8.7G	83.0
CoAtNet-1 [4]	T	42M	8.0G	83.3
CrossFormer-B [32]	T	52M	9.2G	83.4
VMamba-S [20]	S	50M	8.7G	83.6
LocalVMamba-B [16]	S	50M	11.4G	83.7
PlainMamba-L3 [39]	S	50M	14.4G	82.3
GrootV-S [35]	S	51M	8.5G	84.2
QuadMamba-B [36]	S	50M	9.3G	83.8
DefMamba-B	S	51M	8.5G	**84.2**

Table 1. Image classification performance on ImageNet-1K validation set. C, T and S indicate the model type of CNNs, Transformer and SSM. The best results are shown in **bold** font.

Mask R-CNN 1× schedule							
Backbone	**#FLOPs**	**AP^b**	**AP^b_{50}**	**AP^b_{75}**	**AP^m**	**AP^m_{50}**	**AP^m_{75}**
ResNet-50 [12]	260G	38.2	58.8	41.4	34.7	55.7	37.2
Swin-T [21]	267G	42.7	65.2	46.8	39.3	62.2	42.2
ConvNeXt-T [22]	262G	44.2	66.6	48.3	40.1	63.3	42.8
PVTv2-B2 [31]	309G	45.3	67.1	49.4	41.2	64.2	44.4
VMamba-T [20]	286G	47.4	69.5	52.0	42.7	66.3	46.0
LocalVMamba-T [16]	291G	46.7	68.7	50.8	42.2	65.7	45.5
QuadMamba-S [36]	301G	46.7	69.0	51.3	42.4	65.9	45.6
MSVMamba-T [29]	252G	46.9	68.8	51.4	42.2	65.6	45.4
GrootV-T [35]	265G	47.0	69.4	51.5	42.7	66.4	46.0
DefMamba-S	268G	47.5	69.6	51.7	42.8	66.3	46.2

Table 2. Object detection and instance segmentation performance on MSCOCO 2017 val set. All using Mask R-CNN framework. AP^b and AP^m indicate the mean Average Precision (mAP) of detection and segmentation, respectively

ADE20K with crop size 512				
Backbone	**mIOU (SS)**	**mIOU (MS)**	**#Param.**	**#FLOPs**
ResNet-50 [12]	42.1	42.8	67M	953G
Swin-T [21]	44.5	45.8	60M	945G
ConvNeXt-T [22]	46.0	46.7	60M	939G
NAT-T [11]	47.1	48.4	58M	934G
Vim-S [43]	44.9	-	46M	-
LocalVim-S [16]	46.4	47.5	58M	297G
VMamba-T [20]	47.9	48.8	62M	949G
PlainMamba-L2 [39]	46.8	-	55M	285G
LocalVMamba-T [16]	47.9	49.1	57M	970G
MSVMamba-T [29]	47.6	48.5	65M	942G
QuadMamba-S [36]	47.2	48.1	62M	961G
GrootV-T [35]	48.5	49.4	60M	941G
DefMamba-S	48.8	49.6	65M	946G

Table 3. Semantic segmentation performance on ADE20K val set. The crop size is all set to 512^2. SS and MS denote single-scale and multi-scale testing, respectively.

These results demonstrate that deformable scanning translates to improved downstream performance on dense prediction tasks.

4.4. Semantic Segmentation on ADE20K

Setup. We evaluate on the ADE20K [42] benchmark using UperNet [34] with crop size 512 × 512. We follow the training protocol of [20] with 160K iterations.

Results. Table 3 shows semantic segmentation results. **DefMamba-S** achieves **48.8 mIoU** (single-scale) and **49.6 mIoU** (multi-scale), which is the best among SSM-based methods:

- GrootV-T [35]: 48.5/49.4 mIoU (SS/MS);
- VMamba-T [20]: 47.9/48.8 mIoU;
- LocalVMamba-T [16]: 47.9/49.1 mIoU;
- MSVMamba-T [29]: 47.6/48.5 mIoU.

DefMamba-S also surpasses CNN-based ConvNeXt-T [22] (46.0/46.7) and Transformer-based Swin-T [21] (44.5/45.8) by significant margins.

4.5. Efficiency Analysis

Throughput and Latency. Beyond FLOPs, we measure the inference throughput (images/second) of DefMamba on a single NVIDIA A100 GPU with batch size 128 and input resolution 224 × 224. DefMamba-T, -S, and -B

Index	DP	DT	OB	CA	#Param.	#FLOPs	Top-1
(1)					7.5M	1.1G	77.0
(2)	√				7.5M	1.2G	77.4
(3)		√			7.5M	1.2G	77.2
(4)	√	√			7.5M	1.2G	77.9
(5)	√	√	√		8.2M	1.2G	78.2
(6)	√	√	√	√	8.3M	1.2G	**78.6**

Table 4. Comparisons with the different components of proposed deformable scanning in tiny model size settings on ImageNet-1k. The best results are shown in **bold** font.

Index	FB-BB	CB [39]	LB [16]	DB	#Param.	#FLOPs	Top-1
(1)	√				7.3M	1.1G	76.9
(2)				√	8.1M	1.1G	76.5
(3)	√	√			7.5M	1.1G	77.3
(4)	√		√		7.5M	1.1G	77.1
(5)	√			√	8.3M	1.2G	**78.6**

Table 5. Comparisons with the different scanning branches in the setting of tiny model size. DB means our deformable branch. The best results are shown in **bold** font.

achieve throughputs of 1420, 680, and 410 images/sec, respectively. Compared to VMamba-T (630 img/sec) and LocalVMamba-T (590 img/sec) at similar parameter counts, DefMamba-S processes images at 680 img/sec while delivering higher accuracy, confirming that the deformable offset network and gradient-approximated sorting introduce only modest overhead. The tanh-bounded offsets and stage-dependent kernel size schedule together keep this extra cost within 8% of a standard VMamba counterpart at the same model scale, making DefMamba a practical drop-in backbone for latency-sensitive applications.

4.6. Ablation Studies

Components of Deformable Scanning. Table 4 ablates the contribution of each component in DSSM using DefMamba-T on ImageNet-1K. Starting from a baseline with neither deformable points (DP) nor deformable tokens (DT) (row 1, 77.0%):

- Adding **DP alone** (row 2) improves by +0.4% (77.4%);
- Adding **DT alone** (row 3) improves by +0.2% (77.2%);
- Combining **DP+DT** (row 4) yields +0.9% (77.9%), showing synergy;
- Adding **offset bias (OB)** (row 5) further improves to 78.2% (+0.3%);
- Adding **channel attention (CA)** (row 6, full model) achieves **78.6%** (+0.4%).

The final model gains +1.6% over the baseline, confirming the complementarity of all components.

Scanning Branch Design. Table 5 compares different combinations of scanning branches:

- **Forward/Backward (FB-BB) only** (row 1): 76.9% Top-1;
- **Deformable Branch (DB) only** (row 2): 76.5%, lower than FB-BB, showing that the standard branches provide complementary stability;
- **FB-BB + Continuous Branch** [39] (row 3): 77.3%;
- **FB-BB + Local Branch** [16] (row 4): 77.1%;
- **FB-BB + DB** (row 5, ours): **78.6%**, significantly outperforming all alternatives.

The results confirm that combining the stable bidirectional standard branches with the content-adaptive deformable branch yields the best performance. Using the deformable branch alone degrades performance relative to the standard branches, highlighting the importance of the complementary scanning strategy.

Offset Network Kernel Size. We further ablate the stage-dependent depthwise convolution kernel sizes used in the offset network. Using a uniform 3×3 kernel across all four stages yields 78.1% Top-1 on ImageNet-1K, while a uniform 5×5 kernel gives 78.3%. Our stage-dependent schedule $\{9, 7, 5, 3\}$—larger kernels in early stages where spatial resolution is high and smaller kernels in later stages where semantics are richer—achieves the best 78.6%. Intuitively, the offset network in early stages must aggregate contextual features over large neighborhoods to identify coarse object boundaries and guide deformable sampling toward salient regions; later stages require only local adjustments to refine the positioning of semantic tokens. We additionally verify that halving the bottleneck ratio in the channel attention module from $1/4$ to $1/8$ of the channel dimension reduces the offset network parameter count by approximately 15% with only a 0.1% accuracy drop, confirming that the offset prediction module can be made highly lightweight without substantially degrading offset quality.

4.7. Visualization

Fig. 3 visualizes the learned deformable points and reordered token indices for representative images. The deformable sampling points (DP) tend to cluster around object boundaries and semantically salient regions, confirming that the offset network successfully learns to focus on informative areas. The reordered token sequence (DT) groups semantically similar tokens together in the 1D sequence, making the sequential SSM processing more structure-aware.

Fig. 4 shows Grad-CAM-style activation maps at specific token positions. Compared to a standard VMamba baseline, DefMamba produces sharper, more object-centric activations, demonstrating that deformable scanning improves the spatial selectivity of learned representations.

Fig. 5 provides a qualitative comparison of different scanning strategies. Standard raster and window-based

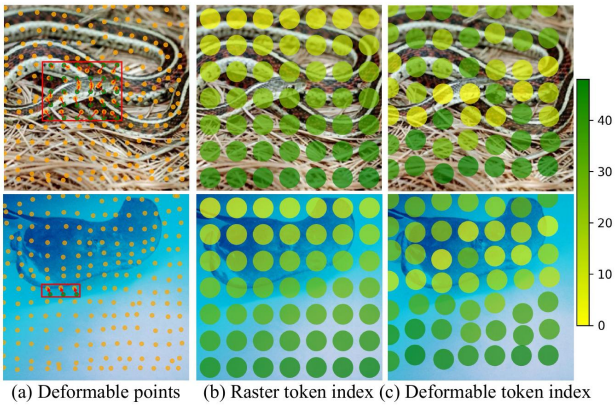

Figure 3. Visualization of deformable points and deformable token indices. The deformable points (left) concentrate on object boundaries and salient regions, while the token reordering (right) clusters semantically similar tokens together in the 1D sequence.

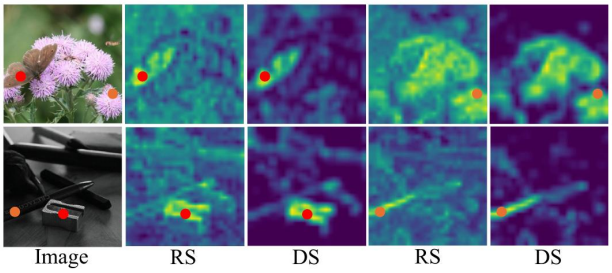

Figure 4. Visualization of activation maps at specific positions. DefMamba's activation maps (right) show sharper, more object-centric responses compared to the standard scanning baseline (left), confirming the benefit of content-adaptive scanning.

scans impose fixed orderings regardless of image content, while DefMamba's deformable scan produces content-adaptive sequences that better respect object boundaries.

5. Conclusion

We have presented **DefMamba**, a deformable visual state space model that addresses the fundamental limitation of fixed, content-agnostic scanning in visual Mamba architectures. By introducing the Deformable State Space Model (DSSM), which jointly learns spatial point offsets and token index offsets, DefMamba adapts the scanning order to image content, aligning 1D SSM sequences with 2D object structures. Key design choices include bounded offsets via tanh normalization for training stability, relative positional bias for position-consistency, gradient approximation through sorting, and complementary forward/backward standard branches.

Extensive experiments on ImageNet-1K, COCO, and ADE20K demonstrate that DefMamba achieves state-of-

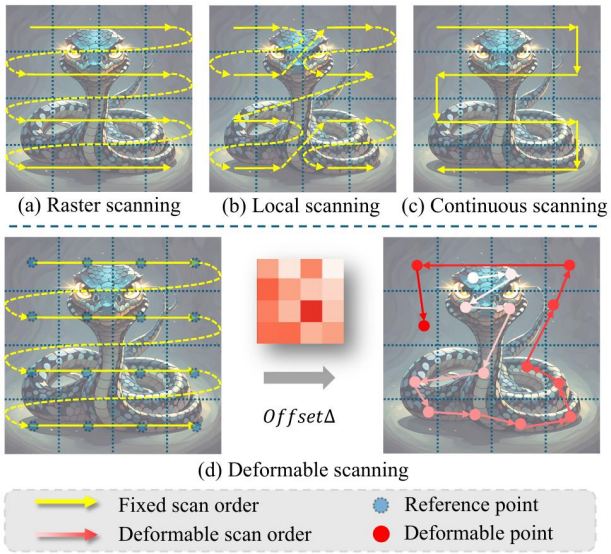

Figure 5. Comparison of deformable scanning and other scanning methods. (a) Standard raster scan, (b) LocalMamba window scan, (c) PlainMamba continuous scan, and (d) DefMamba deformable scan. DefMamba adapts the scan order to image content, aligning the sequence with object structures.

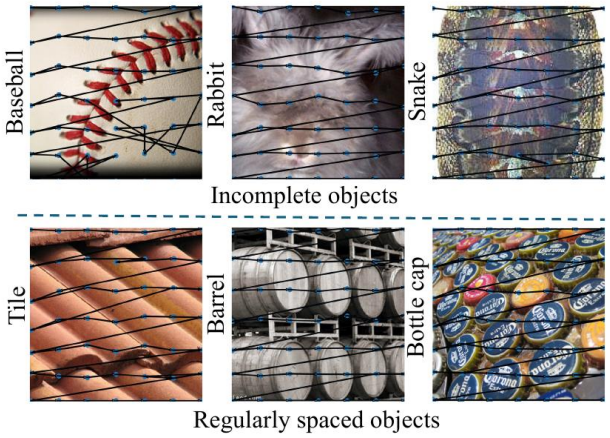

Figure 6. Failure cases of DefMamba. The model can struggle on images with highly irregular textures or low-contrast scenes, where the offset network may not identify a clear object structure to align the deformable scan with.

the-art or competitive results among SSM-based backbones across image classification, object detection, and semantic segmentation, with favorable accuracy/efficiency trade-offs. Ablation studies confirm the contribution of each proposed component, and visualizations validate that the learned deformable scanning indeed focuses on object-relevant regions.

As shown in Fig. 6, DefMamba occasionally struggles on

images with highly irregular or low-contrast textures where the offset network cannot identify a clear structure to align with; addressing such edge cases through uncertainty-aware offset regularization or multi-scale offset supervision is an important direction for future work. Beyond classification and dense prediction, we expect deformable scanning to benefit video understanding and 3D point-cloud processing, where the mismatch between native data topology and fixed 1D serialization is even more pronounced. We believe Def-Mamba opens a promising direction for content-aware sequential processing in visual SSMs, and we hope it will inspire further work on adaptive scanning strategies and their applications to diverse vision tasks.

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